Amagi FMEDA v1.1 — Failure Modes, Effects, and Diagnostic Analysis

VERSION 1.1 — OCTOBER 2026

> Quantitative safety integrity analysis for the AMAGI hardware TCB per IEC 61508-2. Six subsystem groupings analyzed: λ_DU = 18.9 FIT (component-level), SFF = 93% (system-level with CRTM/EGW), PFDavg = 6.9×10^{-5} (1oo1, SIL 2) / 6.9×10^{-6} (1oo2, SIL 3). Software/AI classified as Certain Event (P≈1), excluded from Safety-Critical Path by INV-003 + INV-005 (defined in NIPU v1.1). Aligned with: Framework v2.0, NIPU v1.1, HARA v1.1, Traceability Matrix v1.1, Master Parameter Sheet v2.1, Master Glossary v1.0, S-APL v2.0, Charter v1.2.

Alexey Mikhailovich Burlai AGIAM Technologies Ltd (In Formation) ORCID: 0009-0001-4679-5967
DOI: 10.5281/zenodo.20753410 License: CC BY-NC 4.0 / S-APL v2.0

1. Executive Summary

This Failure Modes, Effects, and Diagnostic Analysis (FMEDA) provides the component-level failure analysis for the AMAGI Architectural Class hardware, conducted in accordance with IEC 61508-2 §7.4.6 and §7.4.9. The FMEDA is a systematic, bottom-up methodology that identifies each hardware component within the Target of Evaluation (TOE), determines its potential failure modes, classifies each failure as safe or dangerous (detectable or undetectable), and computes the aggregate safety metrics required to demonstrate compliance with the target Safety Integrity Level (SIL). This document serves as the quantitative safety evidence supporting the AMAGI safety argument and is intended for submission to certification bodies evaluating the AMAGI architecture against IEC 61508 requirements. The FMEDA quantifies the failure rates in FIT (Failures In Time, where 1 FIT = 1×10−9 failures per hour), the diagnostic coverage (DC) for each subsystem, and the Safe Failure Fraction (SFF) across the entire AMAGI hardware architecture. The reference implementation target is a Xilinx Artix-7 FPGA with 2,500–3,200 LUTs operating at 100 MHz (post-synthesis 142 MHz) and consuming approximately 1.3W. The analysis covers six subsystem components per the FMEDA breakdown aligned with the TCAD reference article: the SCC FSM (core), TTP delay chain, MEM/MAG (BRAM+OTP), Amagi-Q (PQC core), optocoupled I/O, and clock/power isolation.

The overall results demonstrate compliance with SIL 2 requirements under the 1oo1 voting architecture and SIL 3 under the 1oo2 voting architecture. The component-level SFF is 91.6% (system-level SFF = 93% with CRTM/EGW diagnostic uplift per ADR-006/007), which satisfies the ≥90% threshold for Route 2H with Hardware Fault Tolerance (HFT) = 1. The average Probability of Dangerous Failure (PFDavg) for the 1oo1 architecture is 6.9×10^{-5}, which satisfies the SIL 2 architectural constraint per IEC 61508-2:2010 Table 2 (Type B, HFT = 0, SFF = 93%); numerically the value lies within the SIL 3 band of IEC 61508-1:2010 Table 3 (see §7), so the probabilistic requirement is over-satisfied with a margin of more than one decade. The 1oo2 architecture achieves PFDavg = 6.9×10^{-6}, satisfying the SIL 3 architectural constraint (Type B, HFT = 1); numerically the value even exceeds the SIL 4 band (see §7). The component-level dangerous undetected failure rate λdu = 18.9 FIT (system-level ~15.75 FIT after CRTM/EGW uplift) is dominated by the optocoupled I/O (8.4 FIT) and TTP delay chain (4.5 FIT), which together account for approximately 68% of the total component-level dangerous undetected failures.

2. Scope and Applicable Standards

This FMEDA is conducted in accordance with the following international standards, which provide the normative framework for functional safety assessment of electrical, electronic, and

programmable electronic (E/E/PE) safety-related systems. The primary standard governing the FMEDA methodology and acceptance criteria is IEC 61508-2:2010, which defines the requirements for hardware safety integrity of safety-related systems. Specifically, IEC 61508-2 §7.4.6 mandates the identification of all relevant failure modes and the estimation of safe and dangerous failure rates, while §7.4.9 specifies the requirements for diagnostic coverage and the calculation of the Safe Failure Fraction.

The failure rate prediction methodology follows IEC 62380:2004 (Reliability data handbook — Universal model for prediction of reliability on electronic components), which provides component-level base failure rates adjusted for operating conditions including ambient temperature, thermal cycling, and mission profile. IEC 61508-2 Annex A provides the diagnostic coverage reference tables used to assign conservative DC values to each diagnostic mechanism based on the specific technique employed. These three standards together form the complete normative basis for the quantitative analysis presented in this document.

2.1 Applicable Standards

Standard	Year	Scope	Application in FMEDA
IEC 61508-2	2010	Functional safety of E/E/PE systems	FMEDA methodology, SFF/PFD calculation, SIL verification
IEC 62380	2004	Failure rate prediction for electronic components	Component base failure rates (FIT)
IEC 61508-2 Annex A	2010	Diagnostic coverage tables	DC assignment for each diagnostic technique

Table 1: Applicable standards

2.2 TOE Coverage

The FMEDA covers the complete AMAGI Target of Evaluation (TOE), which consists of the following six subsystem groupings per the TCAD reference article: (1) the SCC FSM (core), comprising the Safety Control Core Finite State Machine and its associated control logic; (2) the TTP delay chain, comprising the PVT-calibrated DDL and Access Trigger Layer (ATL) that enforce the Tautological Truth Predicate; (3) MEM/MAG (BRAM+OTP), combining the Secure Hardware Memory Firewall with ECC, the Memory Access Graph OTP array, and the Core Root of Trust for Measurement ROM; (4) the Amagi-Q Post-Quantum Cryptography core; (5) the optocoupled I/O subsystem, comprising the Emergency Gate Watchdog AND-gate, the audit diode, and all galvanically-isolated input/output interfaces; and (6) the clock/power isolation subsystem, comprising the Analog Fallback Circuit, independent LDO, and clock distribution network. All six subsystem groupings are implemented within the Xilinx Artix-7 FPGA fabric, with CRTM realized as a mask-ROM region and MAG OTP realized as one-time programmable fuses or antifuses.

2.3 Exclusions

The following elements are excluded from the FMEDA scope, as permitted by IEC 61508-2 §7.4.2.1 and justified by the AMAGI safety argument. Cognitive clusters (the Normal World processing elements including GEN, COM, ANL, and EXE) are excluded because they reside outside the TOE boundary; the AMAGI invariants do not depend on their correct operation, and the EGW AND-gate ensures that cognitive cluster failure cannot cause dangerous actuation. The power supply is excluded per IEC 61508-2 §7.4.2.1, which allows the assumption of a reliable power supply when the supply is separately certified or when the safety function response to power loss is defined; the AFC provides the defined safe-state response to power loss. The external 2FA signal is assumed to be available as an independent safety input, per the EGW AND-gate design where Actuator_Power = SCC_Enable AND Ext_2FA_Enable.

3. FMEDA Methodology

The FMEDA process follows the systematic procedure defined in IEC 61508-2 §7.4.6, which requires a bottom-up analysis of each hardware component to identify all relevant failure modes, classify them as safe or dangerous, and determine the extent to which dangerous failures can be detected by automatic diagnostic measures. The methodology consists of six sequential steps, each producing a specific output that feeds into the subsequent step. The results of this process form the quantitative basis for demonstrating that the AMAGI hardware architecture achieves the required Safety Integrity Level.

Step 1: Component Identification. Each hardware component within the AMAGI TOE is identified and catalogued, including its function, technology type (combinatorial logic, sequential logic, mixed-signal, analog, OTP, ROM, SRAM), and implementation technology (FPGA LUT, mask-ROM, OTP, discrete analog). This step establishes the complete list of components subject to failure analysis and ensures that no safety-relevant element is omitted from the assessment.

Step 2: Failure Mode Determination. For each component, the possible failure modes are identified based on the component technology and its function within the AMAGI architecture. Each failure mode is classified as one of three categories: safe failure (λsafe), which drives the system toward a safe state; dangerous detectable failure (λdd), which could cause a dangerous condition but is detected by automatic diagnostics within the proof test interval; and dangerous undetectable failure (λdu), which could cause a dangerous condition and is not detected by any automatic diagnostic. This classification is critical because only λdu contributes to the PFDavg calculation.

Step 3: Failure Rate Estimation. The failure rate for each component is estimated in FIT (1 FIT = 1×10−9 failures per hour) using the prediction methodology of IEC 62380:2004. This standard provides base failure rates for electronic component categories (FPGA logic, combinatorial logic, mixed-signal, analog, ROM, OTP, SRAM) adjusted for the specific operating conditions of the AMAGI implementation, including ambient temperature, junction temperature rise, thermal cycling amplitude, and the mission profile of the target application.

Step 4: Diagnostic Coverage Assignment. The diagnostic coverage (DC) for each component is assigned based on the specific diagnostic technique employed, using the reference values in IEC 61508-2 Annex A. The DC represents the fraction of dangerous failures that are detected by the automatic diagnostic measure: DC = λdd / (λdd + λdu). Conservative values are used throughout; where the Annex A tables provide a range, the lower bound is adopted to avoid overstating the safety performance of the architecture.

Step 5: SFF Computation. The Safe Failure Fraction is computed as SFF = (λsafe + λdd) / λtotal, where λtotal = λsafe + λdd + λdu. The SFF represents the fraction of all failures that either drive the system toward a safe state or are detected by automatic diagnostics. A higher SFF indicates greater resilience against dangerous undetected failures and allows a higher SIL to be claimed for a given hardware fault tolerance.

Step 6: SIL Verification. The computed SFF and the hardware fault tolerance (HFT) of the system architecture are verified against IEC 61508-2 Table 2, which specifies the maximum SIL achievable for a given combination of SFF and HFT. The AMAGI architecture supports two voting configurations: 1oo1 (one-out-of-one, HFT=0), where a single dangerous undetected failure causes loss of the safety function; and 1oo2 (one-out-of-two, HFT=1), where two independent dangerous undetected failures are required to cause loss of the safety function. The 1oo2 architecture is achieved through the EGW AND-gate, which requires both the SCC Enable signal and the external 2FA Enable signal to be present for actuation, effectively providing a dual-channel safety barrier.

4. Component Failure Rate Estimates

This section presents the failure rate estimates for each component of the AMAGI TOE. The failure rates are derived from IEC 62380:2004 prediction methodology, calibrated to the operating conditions of the Xilinx Artix-7 FPGA reference implementation. Each component is characterized

by its total failure rate (λtotal), the safe failure rate (λsafe), the dangerous detectable failure rate (λdd), the dangerous undetectable failure rate (λdu), and the diagnostic coverage (DC). The failure mode distribution (safe vs. dangerous) follows the default ratios recommended by IEC 61508-6 for each component type, and the DC values are based on the specific diagnostic techniques implemented in the AMAGI architecture per IEC 61508-2 Annex A.

It is important to note that these are conservative estimates. The IEC 62380 prediction methodology tends to overestimate failure rates for modern FPGA technologies compared to field return data, particularly for devices manufactured on mature process nodes. The Artix-7 family is manufactured on a 28 nm process, which has well-characterized reliability and lower defect density than the assumptions embedded in the IEC 62380 generic models. However, using the standard prediction values provides a defensible, conservative basis for SIL claims.

Component	λ (FIT)	DC (%)	λDU (FIT)	Method
SCC FSM (core)	60	90	3.0	FMEA+FTA
TTP delay chain	40	80	4.5	FMEA+DDL test
MEM/MAG (BRAM+OTP)	25	99	0.2	ECC+OTP verify
Amagi-Q (PQC core)	30	92	1.8	FMEA+signature
Optocoupled I/O	50	70	8.4	Expert+field
Clock/power isolation	20	85	1.0	FMEA+monitor
Component sum (TCAD baseline)	225		18.9	
EGW (AND-gate) see note	5	99	0.05	BIST truth-table test

Table 2: Component failure rate estimates and diagnostic coverage (per TCAD article Table IV)

Note on the EGW (AND-gate) row: The Executive Gateway (EGW) is a non-programmable hardware AND-gate (Actuator_Power = SCC_Enable AND Ext_2FA_Enable) introduced per ADR-006. Its total failure rate of 5.0 FIT (cited in NIPU v1.1 §3.3) is dominated by input/output buffer and metallization failures on the FPGA implementation; the AND function itself is realised by a single LUT with negligible independent failure modes. The 99% diagnostic coverage is achieved by a power-on BIST that exercises the full AND-gate truth table within 60 seconds and by continuous online monitoring of the SCC_Enable and Ext_2FA_Enable signal integrity. The resulting λDU = 0.05 FIT reflects only the rare simultaneous failure of both input paths that escapes the BIST detection window. The EGW is listed here for traceability with NIPU v1.1 §3.3 (which cites 5.0 FIT total failure rate) and to make explicit that the EGW is included in the AMAGI FMEDA scope. However, the “Component sum (TCAD baseline)” row above retains the 225 FIT / 18.9 FIT totals from the TCAD reference article Table IV, which did not separately itemise the EGW; the EGW’s 5 FIT contribution is therefore treated as a conservative disclosure additional to the baseline rather than as a replacement of it. The SFF computation in §6 uses the TCAD baseline (225 FIT / 18.9 FIT) so that the resulting SFF = 91.6% (component) / 93% (system) remains consistent with the values reported in NIPU v1.1, Framework v2.0 §17, OpenSpec v1.1, HARA v1.0 and the NIST AI RMF Profile v1.1. Including the EGW row explicitly would change SFF to 91.8% (component), which does not alter the SIL classification (still ≥90%, SIL 2 with HFT = 0) but would create a documentation inconsistency across the AMAGI suite.

Note on the MEM/MAG (BRAM+OTP) diagnostic coverage: The MEM/MAG subsystem achieves the highest DC (99%) of any subsystem because ECC single-error correction detects all single-bit and double-bit errors in protected BRAM, while OTP read-back verification at boot combined with CRTM Golden Hash comparison detects nearly all dangerous OTP failures prior to operational use. The resulting λDU = 0.2 FIT is the lowest of any subsystem, reflecting the comprehensive detection capability of the combined ECC and OTP verification mechanisms.

5. Diagnostic Coverage Analysis

This section provides a detailed analysis of the diagnostic mechanisms employed for each AMAGI subsystem grouping, justifying the DC values assigned in Table 2. The diagnostic coverage values are derived from IEC 61508-2 Annex A, which provides reference DC values for specific diagnostic techniques. Where multiple diagnostic techniques are applied to a single subsystem, the combined DC is computed conservatively (i.e., the highest single-technique DC is used rather than the theoretical combined coverage, unless the techniques are provably independent). All diagnostic measures are implemented in hardware or firmware within the AMAGI Secure World, ensuring that they cannot be bypassed or disabled by the Normal World cognitive clusters.

5.1 SCC FSM (core) (DC = 90%)

The Safety Control Core FSM is the central sequencing element of the AMAGI architecture, controlling the system state transitions among INIT, IDLE, ARMED, ACTIVE, FAULT, and CUTOFF states. The primary diagnostic mechanism is FMEA (Failure Mode and Effects Analysis) combined with FTA (Fault Tree Analysis), which systematically identifies all failure modes of the SCC FSM and traces their effects through the system. Online FSM state transition monitoring verifies that every state transition follows a legal path as defined by the SCC state transition table; any illegal transition triggers an immediate transition to the FAULT state. Additionally, SPIN model checking of the SCC FSM (25,927,680 reachable states, 15 LTL properties (9 safety + 2 architectural assumptions + 4 liveness)) provides formal verification that the FSM design is free from deadlock, livelock, and assertion violations for the 9 safety properties (zero violations); the 4 liveness properties (SCC-L01 through SCC-L04) produce counterexamples in the unconstrained-environment model. Under the fairness assumptions FA1/FA2, L01–L03 are discharged; the latch-up trap exposed by L04 was closed by the FSM chain-reset correction (NIPU v1.1 §5.1), and the corrected model scc_amagi_v3 verifies L01–L04 with FA1/FA2 (2026-10; verification package per VC v1.1 §9). The DC of 90% is assigned per IEC 61508-2 Annex A for “FMEA+FTA” combined with “online monitoring of program sequence” and “formal verification.” The remaining 10% of dangerous failures are attributed to subtle timing violations that may not cause an illegal state transition but could result in a transient incorrect output.

5.2 TTP Delay Chain (DC = 80%)

The TTP delay chain comprises the PVT-calibrated Digital Delay Lock and the Access Trigger Layer (ATL), which together enforce the Tautological Truth Predicate by requiring that a configurable delay elapses before the actuation command is forwarded. The primary diagnostic mechanism is FMEA combined with a DDL test: the PVT calibration with replica-path feedback tracks the current PVT corner and adjusts the DDL delay accordingly, while the delay bound checking mechanism verifies that the DDL output delay remains within the specified tolerance window during each calibration cycle. Additionally, SPIN model checking of the ATL state machine formally verifies that the ATL cannot be bypassed, shortened, or circumvented. The DC of 80% is assigned per IEC 61508-2 Annex A for “FMEA+DDL test” of a mixed-signal component, which provides lower coverage than purely digital diagnostics because analog drift may escape detection between calibration points. The remaining 20% accounts for PVT transients faster than the calibration cycle and for common-mode drift affecting both the reference chain and the operational chain simultaneously. The TTP delay chain contributes $\lambda DU = 4.5$ FIT, the second-highest dangerous undetected failure rate among the six subsystems.

5.3 MEM/MAG (BRAM+OTP) (DC = 99%)

The MEM/MAG subsystem combines the Secure Hardware Memory Firewall with ECC protection, the Memory Access Graph OTP array, and the Core Root of Trust for Measurement ROM into a single subsystem grouping. This subsystem achieves the highest diagnostic coverage (99%) of any AMAGI subsystem through three complementary mechanisms: (1) ECC single-error correction, which detects and corrects all single-bit errors and detects all double-bit errors in protected

BRAM; (2) OTP read-back verification at boot, which confirms that the OTP programming is consistent with the expected values; and (3) CRTM Golden Hash comparison, where the CRTM computes a SHA-256 hash over the MAG OTP content and compares it against the embedded Golden Hash. The combination of these mechanisms provides near-complete detection of dangerous memory failures, because any bit-level corruption will cause either an ECC detection, a read-back mismatch, or a hash mismatch. The DC of 99% is assigned per IEC 61508-2 Annex A for "ECC+OTP verify" of a memory subsystem. The resulting λDU = 0.2 FIT is the lowest of any subsystem.

5.4 Amagi-Q — PQC Core (DC = 92%)

The Amagi-Q cryptographic engine implements the post-quantum cryptographic suite (FIPS 203 ML-KEM, FIPS 204 ML-DSA, FIPS 205 SLH-DSA) used for audit trail signing, key establishment, and attestation. The primary diagnostic mechanism is FMEA combined with a signature self-test: a known-answer test vector suite exercises the cryptographic algorithms with pre-computed input/output pairs, verifying that the Amagi-Q hardware produces the correct results for each test case. Additionally, a signature verification self-test signs a known message and verifies the signature, confirming end-to-end correct operation of the signing and verification paths. The DC of 92% is assigned per IEC 61508-2 Annex A for "FMEA+signature" self-verification of a programmable logic component. The 8% residual accounts for subtle computational errors that may not be exercised by the test vector suite, such as side-channel leakage that could compromise the cryptographic keys without affecting the functional output.

5.5 Optocoupled I/O (DC = 70%)

The optocoupled I/O subsystem comprises the Emergency Gate Watchdog AND-gate, the unidirectional audit diode, and all galvanically-isolated input/output interfaces. This subsystem has the highest total failure rate (λ = 50 FIT) and the lowest diagnostic coverage (DC = 70%) of any AMAGI subsystem, making it the dominant contributor to dangerous undetected failures (λDU = 8.4 FIT, representing 44% of the total λDU). The diagnostic mechanism is expert judgment combined with field data: expert analysis of the optocoupler failure modes (LED degradation, phototransistor degradation, isolation breakdown) supplemented by field return data from similar optocoupler applications in safety-critical systems. The DC of 70% is assigned per IEC 61508-2 Annex A for "Expert+field" data of a galvanically-isolated I/O subsystem. The 30% residual reflects the inherent difficulty of detecting gradual optocoupler degradation (especially LED CTR degradation) and the limited ability to test the isolation barrier during normal operation. This subsystem represents the primary target for future diagnostic coverage improvement.

5.6 Clock/Power Isolation (DC = 85%)

The clock/power isolation subsystem comprises the Analog Fallback Circuit, independent LDO, and clock distribution network. The diagnostic mechanism is FMEA combined with hardware monitoring: the AFC provides watchdog heartbeat monitoring that activates the hardware bypass when the SCC heartbeat signal is absent for longer than the timeout period, while the independent LDO is monitored for output voltage within specification, and the clock distribution network is monitored for frequency and jitter within tolerance. The AFC operates from a separate power rail with an independent LDO, ensuring that it remains functional even if the main digital power domain fails. The DC of 85% is assigned per IEC 61508-2 Annex A for "FMEA+monitor" of a power/clock isolation subsystem. The 15% residual accounts for common-cause failures affecting both the primary power domain and the AFC independent LDO simultaneously, and for clock distribution failures that may not be detected by the frequency monitor within the monitoring interval.

Subsystem	λ (FIT)	DC (%)	λDU (FIT)	Method
SCC FSM (core)	60	90	3.0	FMEA+FTA
TTP delay chain	40	80	4.5	FMEA+DDL test

Subsystem	λ (FIT)	DC (%)	λDU (FIT)	Method
MEM/MAG (BRAM+OTP)	25	99	0.2	ECC+OTP verify
Amagi-Q (PQC core)	30	92	1.8	FMEA+signature
Optocoupled I/O	50	70	8.4	Expert+field
Clock/power isolation	20	85	1.0	FMEA+monitor
Sum	225		18.9	

Table 3: Diagnostic coverage summary per TCAD article Table IV

6. SFF Calculation

The Safe Failure Fraction (SFF) is the key metric used by IEC 61508 to determine the maximum SIL achievable by a hardware subsystem for a given hardware fault tolerance. The SFF represents the fraction of all failures that either drive the system toward a safe state or are detected by automatic diagnostics, and is defined as SFF = (λsafe + λdd) / λtotal = 1 − λDU / λ. A higher SFF indicates that a larger proportion of dangerous failures are detected, reducing the probability that a dangerous undetected failure will accumulate and cause loss of the safety function. This section computes the component-level and system-level SFF for the AMAGI TOE.

6.1 Component-Level SFF

The component-level aggregate failure rates are obtained by summing the corresponding failure rates across all six subsystem groupings of the AMAGI TOE:

Parameter	Summation	Result (FIT)
λtotal	60 + 40 + 25 + 30 + 50 + 20	225
λDU	3.0 + 4.5 + 0.2 + 1.8 + 8.4 + 1.0	18.9

Table 4: Component-level aggregate failure rate summation

6.2 Component-Level SFF Computation

Applying the SFF formula to the component-level aggregate failure rates:

SFFcomp = 1 − λDU / λtotal = 1 − 18.9 / 225 = 1 − 0.084 = 91.6%

6.3 System-Level SFF with CRTM/EGW Uplift

The component-level SFF of 91.6% does not account for the additional diagnostic coverage provided by the CRTM (Core Root of Trust for Measurement ROM) and EGW (Emergency Gate Watchdog) architectural mechanisms, which operate at the system level rather than at the individual component level. Per ADR-006 and ADR-007, the CRTM boot verification and EGW dual-channel actuation gate provide system-level diagnostic uplift that reduces the effective system-level λDU from 18.9 FIT to approximately 15.75 FIT. This reduction accounts for the fact that the CRTM boot verification detects additional dangerous failures in the MEM/MAG and Amagi-Q subsystems during the boot-time proof test, and the EGW AND-gate provides an independent safety barrier that detects certain dangerous I/O failures before they can cause hazardous actuation.

The system-level SFF is computed as:

SFFsys = 1 − λDU,sys / λtotal = 1 − 15.75 / 225 = 1 − 0.070 = 93%

The system-level SFF of 93% is used throughout the remainder of this document as the conservative SFF figure. Per IEC 61508-2 Table 2, the architectural constraints for Type B components (complex components with software-interacting behavior, which includes all AMAGI FPGA-based subsystems) are as follows: SFF ≥ 90% with HFT = 0 (1oo1 architecture) allows SIL 2; SFF ≥ 90% with HFT = 1 (1oo2 architecture) allows SIL 3. Therefore, the AMAGI architecture

with system-level SFF = 93% meets the SIL 2 architectural constraint under the 1oo1 configuration and the SIL 3 architectural constraint under the 1oo2 voting configuration.

7. PFDavg Computation

7.0 SIL bands (normative citation)

Per IEC 61508-1:2010 Table 3 (low demand mode of operation), the PFDavg bands are:

SIL	PFDavg band	Risk Reduction Factor
2	$10^{-3} < \text{PFDavg} \leq 10^{-2}$	100 to 1 000
3	$10^{-4} < \text{PFDavg} \leq 10^{-3}$	1 000 to 10 000
4	$10^{-5} < \text{PFDavg} \leq 10^{-4}$	10 000 to 100 000

> Correction to v1.0: the bands were printed as «SIL 2: $10^{-3}..10^{-4}$» and «SIL 3: $10^{-4}..10^{-5}$» with a citation of «IEC 61508-3 Clause 7.3» — both positions were erroneous (one-decade shift; the band table resides in Part 1, Table 3).

7.1 1oo1 Architecture (HFT = 0)

The formulas and $\lambda_{DU,sys} = 15.75 \times 10^{-9}$ h^{-1}, T1 = 8 760 h are unchanged from v1.0:

> $\text{PFDavg}(1oo1) = \lambda_{DU,sys} \times T1 / 2 = 15.75 \times 10^{-9} \times 8\,760 / 2 = \mathbf{6.9 \times 10^{-5}}$

The claim:

> The 1oo1 architecture achieves **SIL 2**. The binding constraint is the **architectural constraint** per IEC 61508-2:2010 Table 2 (Type B, SFF = 93% ≥ 90%, HFT = 0 → SIL 2). The computed PFDavg = 6.9×10^{-5} lies more than one decade below the SIL 2 upper band and falls numerically within the SIL 4 band — the probabilistic requirement is therefore over-satisfied with margin ≥ 1 decade, and the claim is deliberately conservative, architecture-limited.

7.2 1oo2 Architecture (HFT = 1)

The formulas and $\beta = 0.10$ are unchanged from v1.0:

> $\text{PFDavg}(1oo2) = \beta \times \lambda_{DU,sys} \times T1 / 2 = \mathbf{6.9 \times 10^{-6}}$

The claim:

> The 1oo2 architecture achieves **SIL 3**. The binding constraint is again the architectural constraint (Type B, SFF ≥ 90%, HFT = 1 → SIL 3). PFDavg = 6.9×10^{-6} numerically exceeds even the SIL 4 band; the SIL 3 claim is conservative and architecture-limited.

7.3 Proof Test Interval and Mission Time (new)

Quantity	Symbol	Value	Role
Proof test interval	T1	8 760 h (1 year)	Enters PFDavg directly; normative in every deployment profile
Mission / design life	T_M	87 600 h (10 years)	Useful-life assumption for constant FIT rates; does not enter PFDavg

> Resolution of the corpus-level “TI 1 year vs 10 years” collision: these are different quantities. All AMAGI documents shall use T1 for the proof test interval and T_M for the design life (aligned with MPS v2.1 §3.3; Framework v2.1 carries a corrigendum).

8. SIL Verification

8.1 1oo1 Verification (HFT = 0)

With the 1oo1 architecture (HFT = 0), the AMAGI hardware achieves SFF = 93% ≥ 90%, which allows a maximum SIL of SIL 2 per IEC 61508-2 Table 2. The PFDavg = 6.9×10^{-5} lies within the SIL 3 band (10^{-4} to 10^{-3}) per IEC 61508-1:2010 Table 3 and therefore over-satisfies the SIL 2 probabilistic requirement by more than one decade; the binding constraint is the architectural constraint (Table 2, Type B, HFT = 0), consistent with the SIL 2 claim for the 1oo1 architecture. To achieve SIL 3, the hardware fault tolerance must be increased to HFT = 1 through the dual-channel voting architecture provided by the EGW AND-gate.

8.2 1oo2 Verification (HFT = 1)

With the 1oo2 architecture (HFT = 1) provided by the EGW AND-gate, the AMAGI hardware achieves SFF = 93% ≥ 90% with HFT = 1, which allows a maximum SIL of SIL 3 per IEC 61508-2 Table 2. The PFDavg = 6.9×10^{-6} over-satisfies the SIL 3 probabilistic requirement by two orders of magnitude (numerically within the SIL 4 band); the binding constraint is again the architectural constraint (Table 2, Type B, HFT = 1), confirming the SIL 3 claim for the 1oo2 architecture.

8.3 Route 2H Compliance

IEC 61508-2 provides two routes for demonstrating compliance with the hardware safety integrity requirements. Route 1H is based solely on the SFF and HFT values of Table 2; Route 2H allows a higher SIL to be claimed if the component has a proven-in-use record demonstrating a sufficiently low dangerous failure rate based on field return data. The AMAGI SIL 2 (1oo1) and SIL 3 (1oo2) claims rest on Route 1H: SFF = 93% with HFT = 0 caps at SIL 2, and SFF = 93% with HFT = 1 reaches SIL 3 (Table 2, Type B). No proven-in-use record exists for the AMAGI-specific implementation; a field-return data campaign (≥ 3 units over one year of operation) is designated for Phase 1b and may later support a Route 2H argument. The formal verification evidence (SPIN model checking) reduces the probability of systematic design errors in the safety-critical state machines, but it is not a substitute for proven-in-use evidence.

Integrity Capability Formula (canon, MPS v2.1 §3.1): **SIL 2 @1oo1 (current) / SIL 3 @1oo2 (Phase 2, 2027)**. The binding constraint in both configurations is the architectural constraint (IEC 61508-2:2010 Table 2, Type B); HARA v1.1 maps hazards onto these two deployment configurations.

9. Conclusions

This FMEDA demonstrates that the AMAGI hardware architecture achieves a level of safety integrity that meets the requirements for SIL 2 under the 1oo1 voting architecture and SIL 3 under the 1oo2 voting architecture, as defined by IEC 61508-2. The analysis has been conducted systematically in accordance with IEC 61508-2 §7.4.6 and §7.4.9, using failure rate predictions per IEC 62380:2004 and diagnostic coverage values derived from IEC 61508-2 Annex A. The results provide a comprehensive, quantitative basis for the AMAGI safety argument and support the certification submission for functional safety compliance.

The key results of the FMEDA are summarized in the following table. These results confirm that the AMAGI hardware architecture satisfies both the architectural constraints and the PFD constraints for SIL 2 (1oo1) and SIL 3 (1oo2). The system-level SFF of 93% incorporates the CRTM/EGW diagnostic uplift per ADR-006/007, which reduces the effective system-level λDU from 18.9 FIT (component-level) to ~15.75 FIT (system-level).

Metric	Value	Requirement	Compliance
SFF (component-level)	91.6%	≥ 90% for Route 2H	PASS

Metric	Value	Requirement	Compliance
SFF (system-level)	93%	≥ 90% for Route 2H + HFT=1	PASS
PFDavg (1oo1)	6.9×10−5	SIL 2 architectural constraint (Table 2, Type B, HFT = 0); PFDavg in SIL 3 band — over-satisfied	PASS
PFDavg (1oo2)	6.9×10−6	SIL 3 architectural constraint (Table 2, Type B, HFT = 1); PFDavg in SIL 4 band — over-satisfied	PASS
Minimum DC	70%	≥ 60% per component	PASS
Maximum DC	99% (MEM/MAG)	N/A	PASS
λDU (component-level)	18.9 FIT	N/A (lower is better)	PASS
λDU (system-level)	~15.75 FIT	N/A (lower is better)	PASS

Table 7: FMEDA results summary — SIL 2 (1oo1) / SIL 3 (1oo2) compliance verification

The analysis further reveals the following key findings regarding the AMAGI hardware architecture. All six subsystem groupings have diagnostic coverage ≥ 70%, with the highest DC (99%) achieved by the MEM/MAG (BRAM+OTP) subsystem through the combination of ECC protection, OTP read-back verification, and CRTM Golden Hash comparison. The optocoupled I/O subsystem has the highest total failure rate (λ = 50 FIT) and the lowest DC (70%), making it the dominant contributor to λDU (8.4 FIT, 44% of component-level λDU). The TTP delay chain is the second-largest contributor (λDU = 4.5 FIT, 24% of component-level λDU). Together, the optocoupled I/O and TTP delay chain account for approximately 68% of all component-level dangerous undetected failures. These two subsystems represent the primary targets for future diagnostic coverage improvement, which could be achieved through additional online monitoring techniques, redundant I/O channels, or improved DDL calibration coverage.

The AMAGI hardware architecture achieves SIL 3 compliance under the 1oo2 voting architecture with a margin of approximately one order of magnitude above the SIL 3 lower bound (PFDavg = 6.9×10−6 vs. the SIL 3 lower bound of 10−5). Under the 1oo1 architecture, the system achieves SIL 2 with PFDavg = 6.9×10−5. The dual-channel EGW AND-gate provides the HFT = 1 necessary for the SIL 3 claim. This margin provides confidence that the SIL 3 claim is robust against moderate increases in failure rates due to environmental stress, aging, or manufacturing variability. The results confirm that the AMAGI Architectural Class hardware is suitable for deployment in safety-critical applications requiring SIL 3 functional safety integrity, including medical implant controllers (IEC 60601-1), industrial safety systems (IEC 61508), and automotive safety elements (ISO 26262 ASIL D equivalent).

Recommendations for future work: (1) Reduce the optocoupled I/O λDU contribution through redundant I/O channels or improved optocoupler monitoring to further increase the SFF margin; (2) Improve the TTP delay chain diagnostic coverage from 80% to ≥ 90% through continuous DDL calibration or dual-delay-chain comparison; (3) Conduct field return data collection for the Artix-7 FPGA implementation to support the Route 2H proven-in-use argument with empirical evidence; (4) Investigate automated online proof testing to reduce the effective proof test interval from 8,760 hours to a shorter interval, further reducing PFDavg; (5) Extend the FMEDA to include common cause failure (CCF) analysis per IEC 61508-6 Annex D, quantifying the beta factor for the 1oo2 architecture with explicit consideration of the DTI, separate PLL, and separate LDO measures.

10. Change Log

Version	Date	Changes
1.0	2026-06	Initial FMEDA: six subsystem groupings, FIT/DC tables, SFF and PFDavg computation, SIL verification

Version	Date	Changes
1.1	2026-10	PFDavg SIL bands corrected per IEC 61508-1:2010 Table 3 (§7.0); SIL claims re-anchored to the architectural constraint (§1, §7–§9, Table 7); T1/T_M distinction fixed (§7.3); terminology canon (ATL/MAG/CRTM); L04 status synchronized with CCR-02 (§5.1); Route 2H statement corrected (§8.3); references added (§12); abstract SFF aligned to the computed 93%

11. References

1. IEC 61508-1:2010, Functional safety of electrical/electronic/programmable electronic safety-related systems — Part 1: General requirements (Table 3: PFDavg bands, low-demand mode).
2. IEC 61508-2:2010, Functional safety of electrical/electronic/programmable electronic safety-related systems — Part 2: Requirements for electrical/electronic/programmable electronic safety-related systems (Table 2: architectural constraints, Type A/Type B; Annex A: diagnostic coverage).
3. IEC 61508-6:2010, Functional safety of electrical/electronic/programmable electronic safety-related systems — Part 6: Guidelines on the application of IEC 61508-2 and IEC 61508-3 (Annex D: common-cause and beta factors).
4. IEC 62380:2004, Reliability data handbook — Universal model for reliability prediction of electronics components, PCBs and equipment.
5. Burlai, A.M. (2026). AMAGI: A Formal Verification Framework for Hardware-Enforced AI Safety (IEEE TCAD submission TCAD-2026-1011; component failure-rate and DC baseline, Table IV). Rejected 2026-09-22; new submission in preparation.
6. Burlai, A.M. (2026). Amagi Framework Specification v2.0. DOI: 10.5281/zenodo.20753400.
7. Burlai, A.M. (2026). Amagi-NIPU v1.1. DOI: 10.5281/zenodo.20753401.
8. Burlai, A.M. (2026). Amagi Open Specification v1.1. DOI: 10.5281/zenodo.20753403.
9. Burlai, A.M. (2026). Amagi HARA v1.1. DOI: 10.5281/zenodo.20753408 (new version).
10. Burlai, A.M. (2026). AMAGI Master Parameter Sheet v2.1. DOI: 10.5281/zenodo.20145212 (new version).
11. Burlai, A.M. (2026). AMAGI Traceability Matrix v1.1. DOI: 10.5281/zenodo.20753412 (new version).
12. Burlai, A.M. (2026). AMAGI Master Glossary v1.0 (new record — DOI upon publication).
13. Burlai, A.M. (2026). Amagi-SAPL v2.0. DOI: 10.5281/zenodo.20753406.

Supersession note: the PFDavg / SFF / subsystem-type figures of MPS v2.0 §3 are withdrawn; the normative source of the quantitative reliability parameters of the AMAGI TCB is this document (v1.1).